Section Report 2 — WGS: Reads to Info

BIOL 4810

2026-03-12

What you are doing: You are given **raw reads only.** You must run the full WGS workflow and document it in a way that is reproducible and ready to commit to Git.
Submission: Fill this page out, upload the requested screenshots/PDFs into the page, then use **Download as PDF (Print).**
Notes: You should taking detailed notes about each step, what software you are using, what version of the software (if available), and output files so that someone else could theoretically repeat your exact work.

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0) Required folder structure (make this first)

Inside your course repository, create a folder named something like:

```
sectionreport2_wgs/
```

Inside that folder, create:

```
rawfastqc/
fastp/
trimmedfastqc/
assembly/
quast/
filteredassembly/
checkm2/
prokka/
ani/
tygs/
miningfiles/
```

---

1) FASTQC on raw reads (Galaxy)

In Galaxy 1. Upload your raw reads (make sure to create the paired data set) 2. Run **FastQC** on **raw** reads 3. Open the FastQC HTML report

Download requirement: Download the FastQC HTML report for **either R1 or R2** and save it in: `rawfastqc/`

1A. FastQC version (from Galaxy Job Info):

FastQC Galaxy Version 0.74+galaxy1

1B. Read length reported:

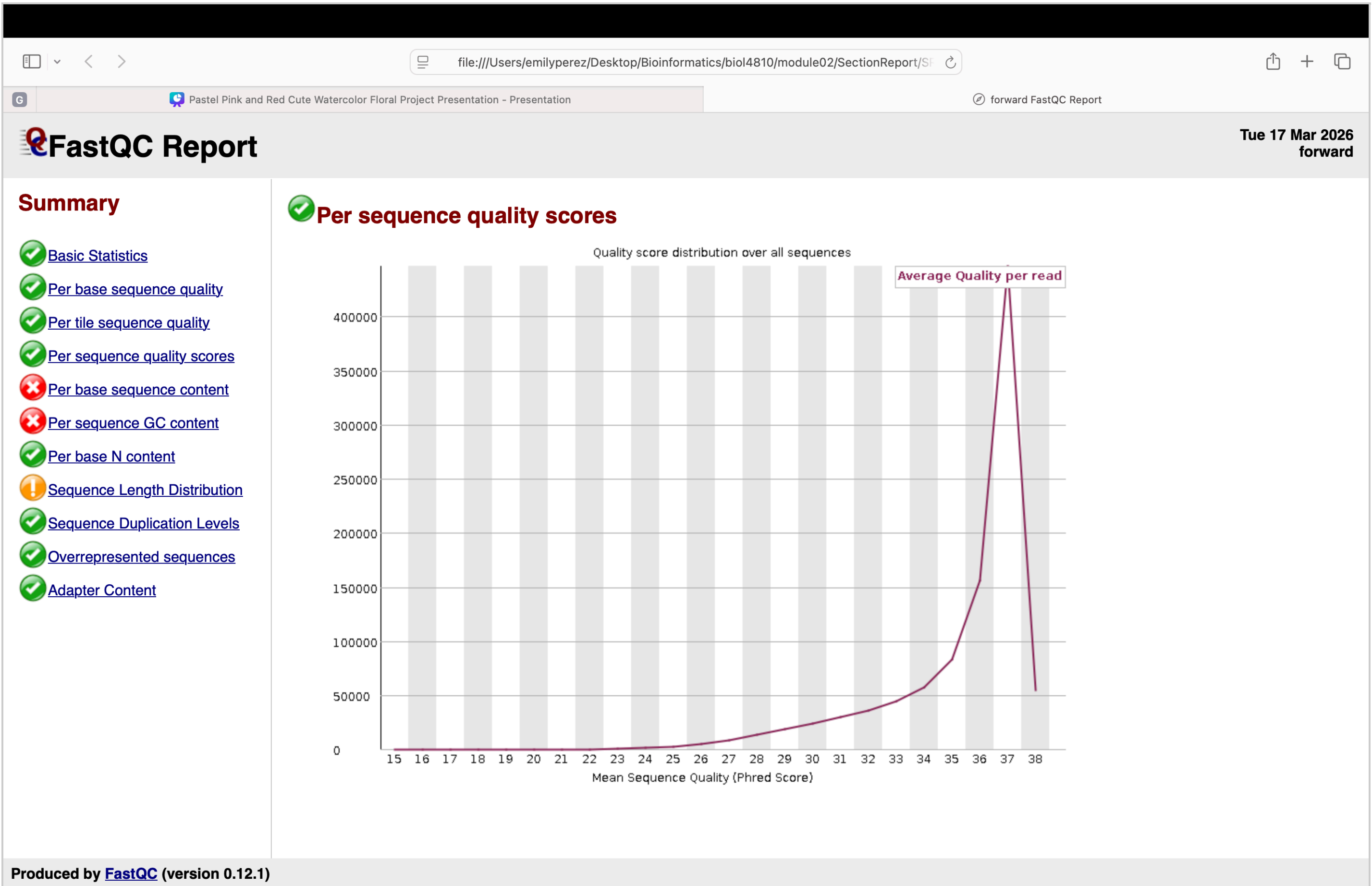
20-251

1C. One key quality issue you see in raw reads (2–3 sentences):

The "Per sequence GC content" is marked as a quality issue. This probably means that the report observed GC distribution and noticed that does not match the expected normal distribution.

Upload screenshot: raw FastQC quality plot (R1 or R2)

Choose File Screensho….12.42 AM



Save the downloaded FastQC HTML report in `rawfastqc/` using your naming convention.

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2) fastp trimming/filtering (Galaxy)

In Galaxy 1. Run **fastp** on your raw reads 2. Keep defaults unless your instructor specifies otherwise 3. Make sure the output includes the report

Download requirement: Download the **fastp report HTML** and save it in: `fastp/`

2A. fastp version (from Galaxy Job Info):

fastp Galaxy Version 1.1.0+galaxy0

2B. What did fastp change? (2–3 sentences)

Fastp removed low-quality bases, trimmed away adapter sequences that it automatically detected, and filtered out reads that were too short. This allows for a cleaner dataset.

2C. Name one metric that changed from BEFORE → AFTER in the fastp report (e.g., Q30%, GC%, read length distribution) and state how it changed:

The total bases got shortened from 445.628487 M to 443.183135 M. It got shorter.

Upload your fastp HTML report (downloaded from Galaxy)

Choose File SR2fastp.html

Selected file: SR2fastp.html

Save your fastp HTML in `fastp/` using your naming convention.

3) FASTQC on trimmed reads (Galaxy)

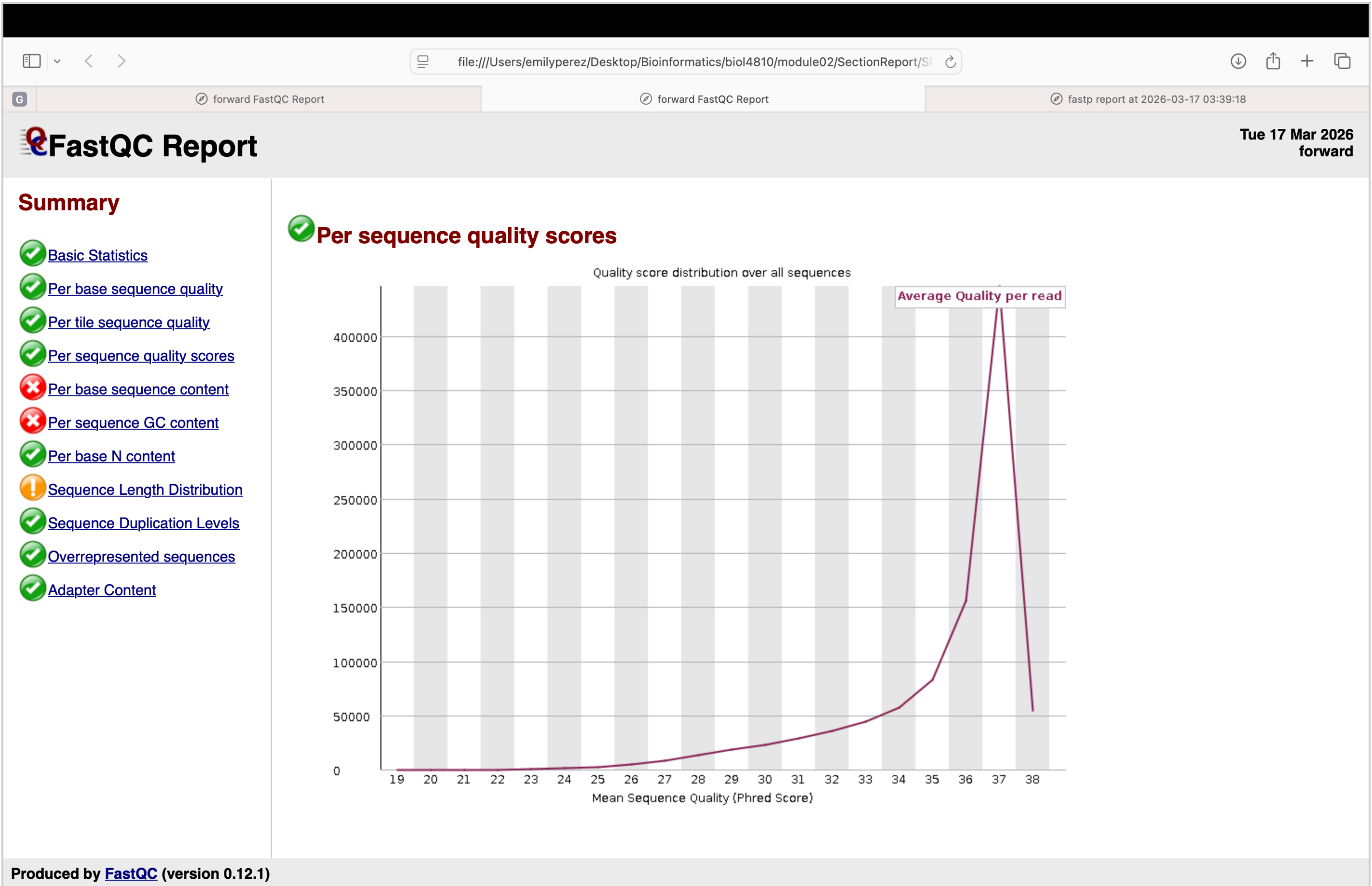
In Galaxy 1. Run **FastQC** on the **trimmed** reads 2. Download the FastQC HTML report for either R1 or R2 and save it in: `trimmedfastqc/`

3A. One key improvement (or remaining issue) after trimming (2–3 sentences):

A remaining issue that appears is the "per sequence GC content." The actual GC distribution still deviates from the expected normal bell curve.

Upload screenshot: trimmed FastQC quality plot (R1 or R2)

Choose File Screensho….45.50 PM



Save the downloaded FastQC HTML report in `trimmedfastqc/` using your naming convention.

4) Assembly (SPAdes in Galaxy)

In Galaxy 1. Run **SPAdes** on your trimmed reads 2. Download the **contigs FASTA** and save it in: `assembly/`

4A. SPAdes version (from Galaxy Job Info):

SPAdes Galaxy Version 4.2.0+galaxy0

4B. Name of your downloaded contigs file (after you renamed it):

SR2assembly_contigs.fasta

5) Assembly evaluation (QUAST in Galaxy)

In Galaxy 1. Run **QUAST** on your SPAdes contigs 2. Download the **QUAST PDF report only** and save it in: `quast/`

5A. QUAST version (from Galaxy Job Info):

Quast Galaxy Version 5.3.0+galaxy1

5B. N50:

571232

5C. L50:

4

5D. Largest contig:

818560

5E. Total assembly length:

5089729

Upload your QUAST PDF report (downloaded from Galaxy)

Choose File SR2Quast_…ntigs.pdf

Selected file: SR2Quast_assembly_contigs.pdf

Save the QUAST PDF in `quast/` using your naming convention.

5F. Based on QUAST, is your assembly fragmented? Use at least 2 metrics to justify (2–4 sentences):

The assembly is not very fragmented and actually looks solid. One big reason is the N50 value of 571,232 bp, which shows that most of the genetic information is grouped into very large chunks. Having a L50 of 4, meaning it only takes four of those large pieces to cover half of the entire genome. Overall, having only 26 contigs for a 5.09 Mbp genome suggests the assembly is good.

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6) Genome quality (CheckM2 in Galaxy)

6A) Filter FASTA (Galaxy)

To reduce tiny contigs, create a filtered assembly for work going forward:

In Galaxy 1. Use **Filter FASTA** 2. Input: your SPAdes contigs FASTA 3. Filter by minimum contig length: **500 bp** 4. Save the filtered contigs as a new dataset in `filteredcontigs`

6A. Confirm your filter rule (one line):

Filtered on sequence length; minimum length 500 bp

6B) In Galaxy

1. Run **CheckM2** on your assembly (contigs)
2. Download the CheckM2 report/output and save it in: `checkm2/`

6B. CheckM2 version (from Galaxy Job Info):

Galaxy Version 1.1.0+galaxy0

6C. Completeness (%):

97.48%

6D. Contamination (%):

0.21%

6E. Interpret these numbers: is this "good enough" for downstream annotation? Why? (2–3 sentences)

This assembly is "good enough" for downstream annotation. It shows completeness of 97.48% and a low contamination of 0.21%. This indicates that nearly the entire genome is present with almost no outside interference.

Folder requirement: Put the downloaded CheckM2 report/output in `checkm2/` with your naming convention.

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7) Prokka annotation (Galaxy)

In Galaxy 1. Run **Prokka** 2. Input: your **filtered** contigs (≥500 bp) 3. Change at least one setting from default as instructed (record it below) 4. Run Prokka

7A. Prokka version (from Galaxy Job Info):

Galaxy Version 1.14.6+galaxy1

7B. What did you change from Prokka defaults? (1–2 sentences)

The only things we changed from default is "Force GenBank compliance- YES".

Download requirements (Prokka): download and save these into `prokka/` - `.txt` (summary) - `.gbk` (GenBank format annotation) - `.ffn` (nucleotide sequences of features)

7C. Paste your Prokka summary lines (from the .txt) and describe why this does or does not look like a typical bacterial genome:

CDS: 4661
gene: 4754
rRNA: 8
tRNA: 84
tmRNA: 1
Yes it looks like a typical bacterial genome. It falls in to the range of 3-6 Mb, 3500-5500 CDS, and 30-100

7D. How many hypothetical proteins does your organism have? What percent of CDS are hypothetical?

24.48% of CDS are hypothetical

rRNA feature check (from Prokka outputs)

Open your Prokka outputs and locate the rRNA entries (typically in the `.ffn` and/or `.tbl` / `.gff` depending on Galaxy wrapper).

7E. Paste the description/header lines for 23S, 16S, and 5S rRNA (three lines total):

>LOECGNNE_04749 23S ribosomal RNA
>LOECGNNE_04751 16S ribosomal RNA
>LOECGNNE_01990 5S ribosomal RNA

7F. Paste the full 16S rRNA FASTA sequence (header + sequence):

ATGCAACGCGAAGAACCTTACCTACTCTTGACATCCAGAGAACTTAGCAGAGATGCTTTG
GTGCCTTCGGGAACTCTGAGACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTTGTGAAAT
GTTGGGTTAAGTCCCGCAACGAGCGCAACCCTTATCCTTTGTTGCCAGCGGTTCGGCCGG
GAACTCAAAGGAGACTGCCAGTGATAAACTGGAGGAAGGTGGGGATGACGTCAAGTCATC
ATGGCCCTTACGAGTAGGGCTACACACGTGCTACAATGGCATATACAAAGAGAAGCGACC
TCGCGAGAGCAAGCGGACCTCATAAAGTATGTCGTAGTCCGGATTGGAGTCTGCAACTCG
ACTCCATGAAGTCGGAATCGCTAGTAATCGTAGATCAGAATGCTACGGTGAATACGTTCC
CGGGCCTTGTACACACCGCCCGTCACACCATGGGAGTGGGTTGCAAAAGAAGTAGGTAGC
TTAACCTTCGGGAGGGCGCTTACCACTTTGTGATTCATGACTGGGGTGAAGTCGTAACAA
GGTAACCGTAGGGGAACCTGCGGTTGGATCACCTCCTT

8) 16S species ID using BLAST (NCBI)

Use your 16S rRNA FASTA from above.

8A. Top hit species name + percent identity + query coverage:

99.41% identity
100% query coverage

8B. Based on 16S alone, how confident are you in species-level ID? Why? (2–3 sentences)

While 16S rRNA sequencing is good for identifying the genus, I have semi-low confidence in my species-level ID because many closely related species share nearly identical 16S sequences. It is possible that I have another species.

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9) ANI (EZBioCloud) — two comparisons

Use the ANI tool and compare your assembly to **two reference genomes** that you choose (based on your best guess from 16S/Prokka/TYGS). Open the site here (new tab): EZBioCloud ANI (https://www.ezbiocloud.net/tools/ani)

Download requirement: download the two reference genomes you used and save them in `ani/` with clear names.

9A. Reference genome #1 (species/strain):

Serratia marcescens

9B. ANI % vs reference #1:

95.43%

9C. Reference genome #2 (species/strain):

Serratia liquefaciens

9D. ANI % vs reference #2:

83.46%

9E. Interpret your ANI: do you meet a typical same-species ANI threshold? What conclusion do you draw? (2–4 sentences)

The threshold for defining a species is an ANI of ≥95%. Since the genome I uploaded has 95.43% ANI with Serratia marcescens, it meets the criteria for being classified as the same species. On the other hand, the 83.46% ANI with Serratia liquefaciens is below the threshold, confirming that my strain is not S. liquefaciens.

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10) TYGS phylogeny (use filtered contigs for this report)

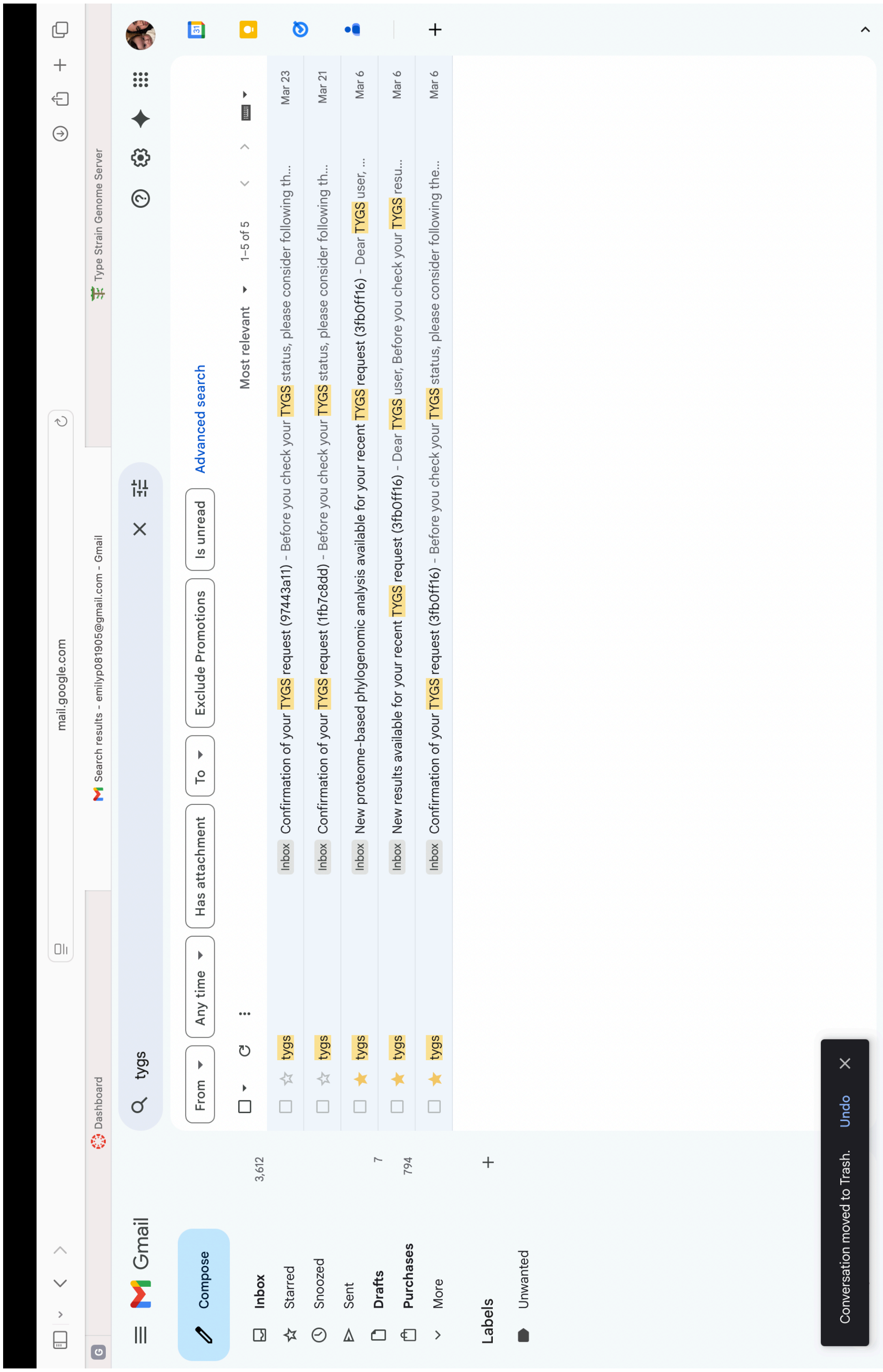
Run TYGS using your filtered contigs (≥500 bp). Export/download the tree (.phy file) and view in iTOL. Make sure that you have displayed the bootstrap values. Export your tree as a .pdf file. The .phy and .pdf should be in your `tygs/` folder

In your own words: what kinds of evidence does TYGS use to place your genome? Did you need to generate a proteome tree or was the genome tree sufficient? Did it correspond to what you found with your 16S BLAST results.

I was unfortunately unable to get the phylogenetic tree through TYGS.

Upload your TYGS/iTOL tree as a PDF

Choose File Tree.pdf



11) Genome mining (Day 14 web tools) — ALL REQUIRED

You must run **all** tools below and include a **screenshot of results** for each. Save screenshots in: `miningfiles/` with clear names.

Also: record the **version/date/database** if the tool reports it (many do on the results page or in the footer).

11A) CARD / RGI (AMR genes)

Open: CARD / RGI (https://card.mcmaster.ca/analyze/rgi)

11A1. CARD/RGI version + database/date info (copy from results page if shown):

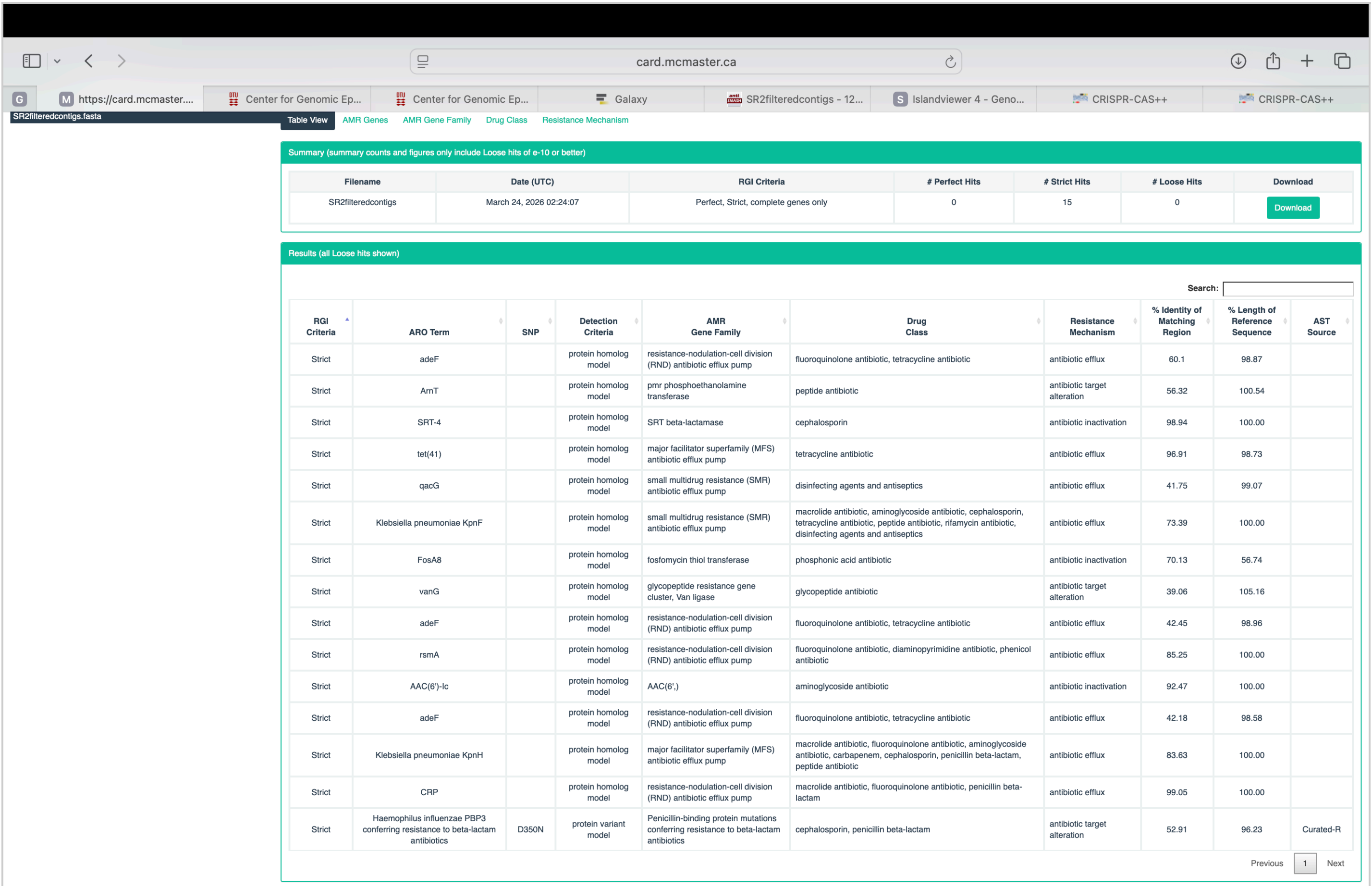
CARD v4.0.1

11A2. Key AMR findings (2–4 sentences). Include at least one gene + evidence type (e.g., perfect/strict/loose):

The first hit was gene "adeF" with strict RGI criteria. Another gene is "SRTSRT-4" with a strict RGI criteria. I don't see any perfect or loose findings.

Upload screenshot: CARD/RGI results

Choose File SR2CARDresults.png



card.mcmaster.ca

SR2filteredcontigs.fasta

Table View | AMR Genes | AMR Gene Family | Drug Class | Resistance Mechanism

Summary (summary counts and figures only include Loose hits of e-10 or better)

Filename	Date (UTC)	RGI Criteria	# Perfect Hits	# Strict Hits	# Loose Hits	Download
SR2filteredcontigs	March 24, 2026 02:24:07	Perfect, Strict, complete genes only	0	15	0	Download

Results (all Loose hits shown)

Search:

RGI Criteria	ARO Term	SNP	Detection Criteria	AMR Gene Family	Drug Class	Resistance Mechanism	% Identity of Matching Region	% Length of Reference Sequence	AST Source
Strict	adeF		protein homolog model	resistance-nodulation-cell division (RND) antibiotic efflux pump	fluoroquinolone antibiotic, tetracycline antibiotic	antibiotic efflux	60.1	98.87	
Strict	ArnT		protein homolog model	pmr phosphoethanolamine transferase	peptide antibiotic	antibiotic target alteration	56.32	100.54	
Strict	SRT-4		protein homolog model	SRT beta-lactamase	cephalosporin	antibiotic inactivation	98.94	100.00	
Strict	tet(41)		protein homolog model	major facilitator superfamily (MFS) antibiotic efflux pump	tetracycline antibiotic	antibiotic efflux	96.91	98.73	
Strict	qacG		protein homolog model	small multidrug resistance (SMR) antibiotic efflux pump	disinfecting agents and antiseptics	antibiotic efflux	41.75	99.07	
Strict	Klebsiella pneumoniae KpnF		protein homolog model	small multidrug resistance (SMR) antibiotic efflux pump	macrolide antibiotic, aminoglycoside antibiotic, cephalosporin, tetracycline antibiotic, peptide antibiotic, rifamycin antibiotic, disinfecting agents and antiseptics	antibiotic efflux	73.39	100.00	
Strict	FosA8		protein homolog model	fosfomycin thiol transferase	phosphonic acid antibiotic	antibiotic inactivation	70.13	56.74	
Strict	vanG		protein homolog model	glycopeptide resistance gene cluster, Van ligase	glycopeptide antibiotic	antibiotic target alteration	39.06	105.16	
Strict	adeF		protein homolog model	resistance-nodulation-cell division (RND) antibiotic efflux pump	fluoroquinolone antibiotic, tetracycline antibiotic	antibiotic efflux	42.45	98.96	
Strict	rsmA		protein homolog model	resistance-nodulation-cell division (RND) antibiotic efflux pump	fluoroquinolone antibiotic, diaminopyrimidine antibiotic, phenicol antibiotic	antibiotic efflux	85.25	100.00	
Strict	AAC(6')-Ic		protein homolog model	AAC(6',)	aminoglycoside antibiotic	antibiotic inactivation	92.47	100.00	
Strict	adeF		protein homolog model	resistance-nodulation-cell division (RND) antibiotic efflux pump	fluoroquinolone antibiotic, tetracycline antibiotic	antibiotic efflux	42.18	98.58	
Strict	Klebsiella pneumoniae KpnH		protein homolog model	major facilitator superfamily (MFS) antibiotic efflux pump	macrolide antibiotic, fluoroquinolone antibiotic, aminoglycoside antibiotic, carbapenem, cephalosporin, penicillin beta-lactam, peptide antibiotic	antibiotic efflux	83.63	100.00	
Strict	CRP		protein homolog model	resistance-nodulation-cell division (RND) antibiotic efflux pump	macrolide antibiotic, fluoroquinolone antibiotic, penicillin beta-lactam	antibiotic efflux	99.05	100.00	
Strict	Haemophilus influenzae PBP3 conferring resistance to beta-lactam antibiotics	D350N	protein variant model	Penicillin-binding protein mutations conferring resistance to beta-lactam antibiotics	cephalosporin, penicillin beta-lactam	antibiotic target alteration	52.91	96.23	Curated-R

Previous 1 Next

Save as something like `SampleID_card_rgi.png` in `miningfiles/` .

11B) ResFinder (AMR genes)

Open: ResFinder (https://genepi.food.dtu.dk/resfinder)

11B1. ResFinder version/database info (copy from results page if shown):

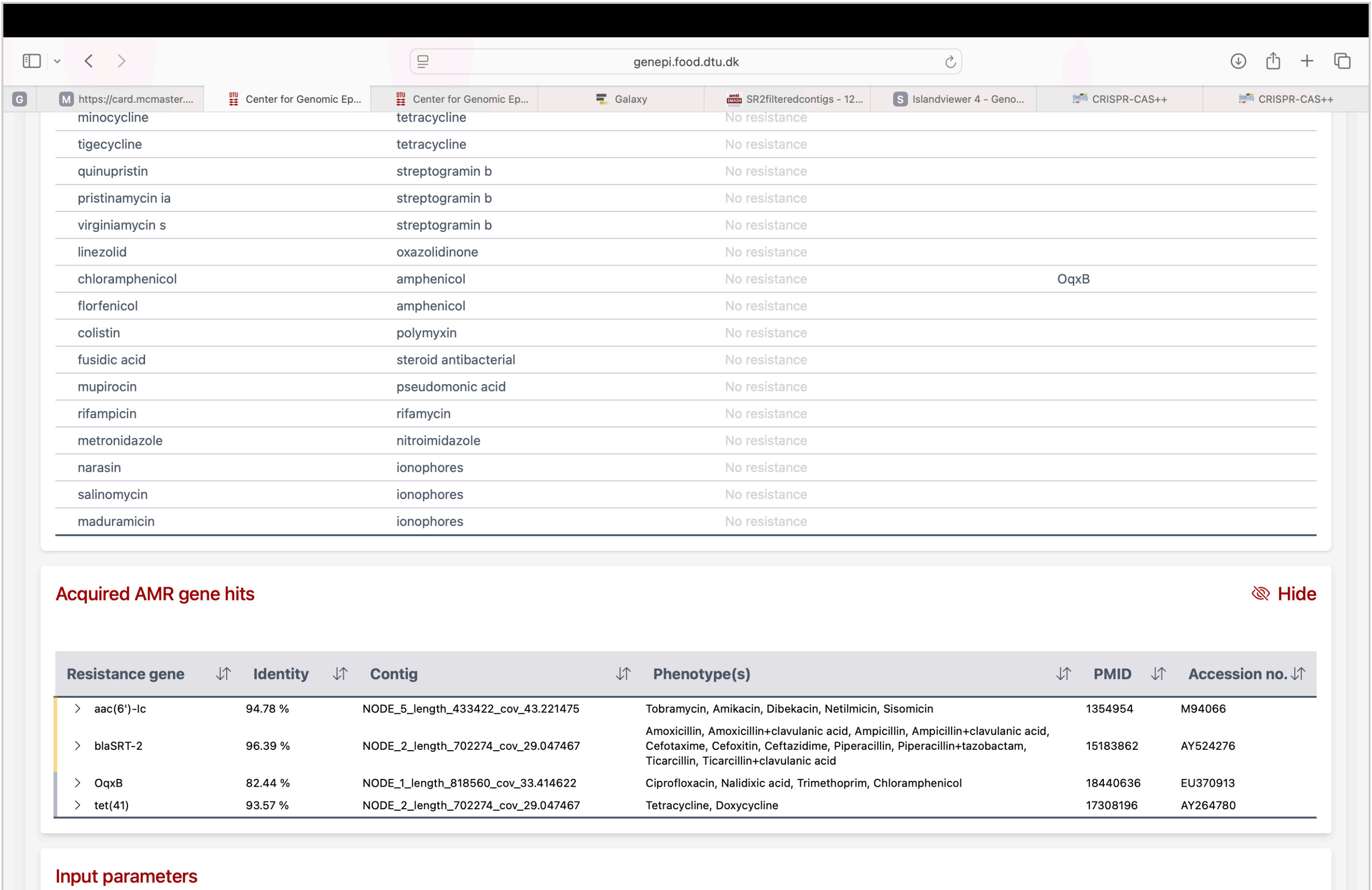
ResFinder-4.7.2

11B2. Key findings (2–4 sentences). Do ResFinder and CARD agree? If not, why might they differ?

Both ResFinder and CARD successfully identified the aac(6')-Ic gene. This is a specific resistance marker. ResFinder focuses primarily on acquired genes found, while CARD includes a broader range that can include chromosomal mutations.

Upload screenshot: ResFinder results

Choose File SR2ResFin…sults.png



minocycline	tetracycline	No resistance	
tigecycline	tetracycline	No resistance	
quinupristin	streptogramin b	No resistance	
pristinamycin ia	streptogramin b	No resistance	
virginiamycin s	streptogramin b	No resistance	
linezolid	oxazolidinone	No resistance	
chloramphenicol	amphenicol	No resistance	OqxB
florfenicol	amphenicol	No resistance	
colistin	polymyxin	No resistance	
fusidic acid	steroid antibacterial	No resistance	
mupirocin	pseudomonic acid	No resistance	
rifampicin	rifamycin	No resistance	
metronidazole	nitroimidazole	No resistance	
narasin	ionophores	No resistance	
salinomycin	ionophores	No resistance	
maduramicin	ionophores	No resistance	

Acquired AMR gene hits

Hide

Resistance gene	Identity	Contig	Phenotype(s)	PMID	Accession no.
aac(6')-Ic	94.78 %	NODE_5_length_433422_cov_43.221475	Tobramycin, Amikacin, Dibekacin, Netilmicin, Sisomicin	1354954	M94066
blaSRT-2	96.39 %	NODE_2_length_702274_cov_29.047467	Amoxicillin, Amoxicillin+clavulanic acid, Ampicillin, Ampicillin+clavulanic acid, Cefotaxime, Cefoxitin, Ceftazidime, Piperacillin, Piperacillin+tazobactam, Ticarcillin, Ticarcillin+clavulanic acid	15183862	AY524276
OqxB	82.44 %	NODE_1_length_818560_cov_33.414622	Ciprofloxacin, Nalidixic acid, Trimethoprim, Chloramphenicol	18440636	EU370913
tet(41)	93.57 %	NODE_2_length_702274_cov_29.047467	Tetracycline, Doxycycline	17308196	AY264780

Input parameters

Save as `SampleID_resfinder.png` in `miningfiles/` .

11C) PathogenFinder 2.0 (pathogenicity prediction)

Open: PathogenFinder (https://genepi.food.dtu.dk/pathogenfinder)

11C1. PathogenFinder version/database info (copy from results page if shown):

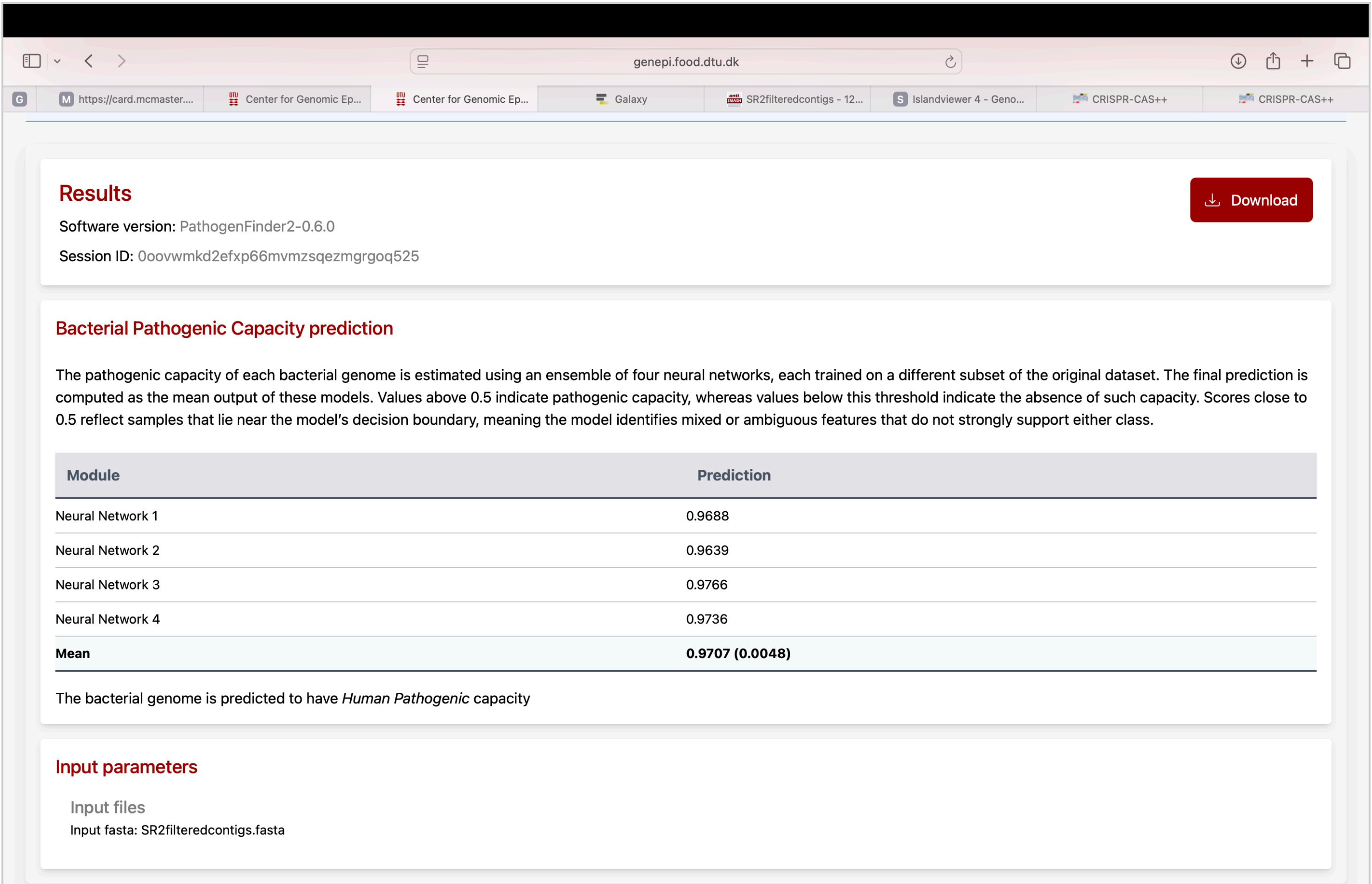
PathogenFinder2-0.6.0

11C2. Report the predicted probability/label and interpret it (2–4 sentences):

The PathogenFinder score of 0.9707 indicates a very high probability that the bacterial isolate is a human pathogen. This score is significantly above the standard threshold (usually 0.5), leading PathogenFinder to officially label the genome as having "Human Pathogenic" capacity.

Upload screenshot: PathogenFinder results

Choose File SR2Pathog…sults.png



genepi.food.dtu.dk

https://card.mcmaster... | Center for Genomic Ep... | Center for Genomic Ep... | Galaxy | SR2filteredcontigs - 12... | Islandviewer 4 - Geno... | CRISPR-CAS++ | CRISPR-CAS++

Results

Download

Software version: PathogenFinder2-0.6.0

Session ID: 0oovwmkd2efxp66mvmzsqezmgrgoq525

Bacterial Pathogenic Capacity prediction

The pathogenic capacity of each bacterial genome is estimated using an ensemble of four neural networks, each trained on a different subset of the original dataset. The final prediction is computed as the mean output of these models. Values above 0.5 indicate pathogenic capacity, whereas values below this threshold indicate the absence of such capacity. Scores close to 0.5 reflect samples that lie near the model's decision boundary, meaning the model identifies mixed or ambiguous features that do not strongly support either class.

Module	Prediction
Neural Network 1	0.9688
Neural Network 2	0.9639
Neural Network 3	0.9766
Neural Network 4	0.9736
Mean	**0.9707 (0.0048)**

The bacterial genome is predicted to have *Human Pathogenic* capacity

Input parameters

Input files

Input fasta: SR2filteredcontigs.fasta

Save as `SampleID_pathogenfinder.png` in `miningfiles/` .

11D) VFDB virulence factors via **ABRicate** (Galaxy)

The VirulenceFinder/VFDB web outputs can be inconsistent, so for this report you will run VFDB searches **inside Galaxy** using **ABRicate**.

In Galaxy (ABRicate): 1. Search tools for **ABRicate** 2. **Input:** your **Prokka GenBank file** (`.gbk`) 3. Under**Advanced**, **Database:** choose **VFDB** 4. Run ABRicate 5. Download the **tabular** results file (`.tabular`) and save it in `miningfiles/` using your naming convention.

11D1. ABRicate version + VFDB database version (from Galaxy Job Info):

ABRicate (Galaxy Version 1.0.1)

11D2. How many VFDB hits did you get (from the results table/summary)?

3

11D3. List your top 3 VFDB hits (gene name + brief description/function). If you have fewer than three, list all. If you have none, you can put NA:

Gene 1: flgH encodes the flagellar L-ring protein
Gene 2: fliM encodes for the motor switch protein FliM
Gene 3: fliG encodes for the moro switch protein G

Upload your ABRicate VFDB results table (tabular)

Choose File SR2ABRicat…s.tabular

Selected file: SR2ABRicateResults.tabular

Save as `SampleID_abricate_vfdb.tabular` in `miningfiles/`.

11E) antiSMASH (biosynthetic gene clusters)

Open: antiSMASH (https://antismash.secondarymetabolites.org/)

11E1. antiSMASH version/database info (copy from results page if shown):

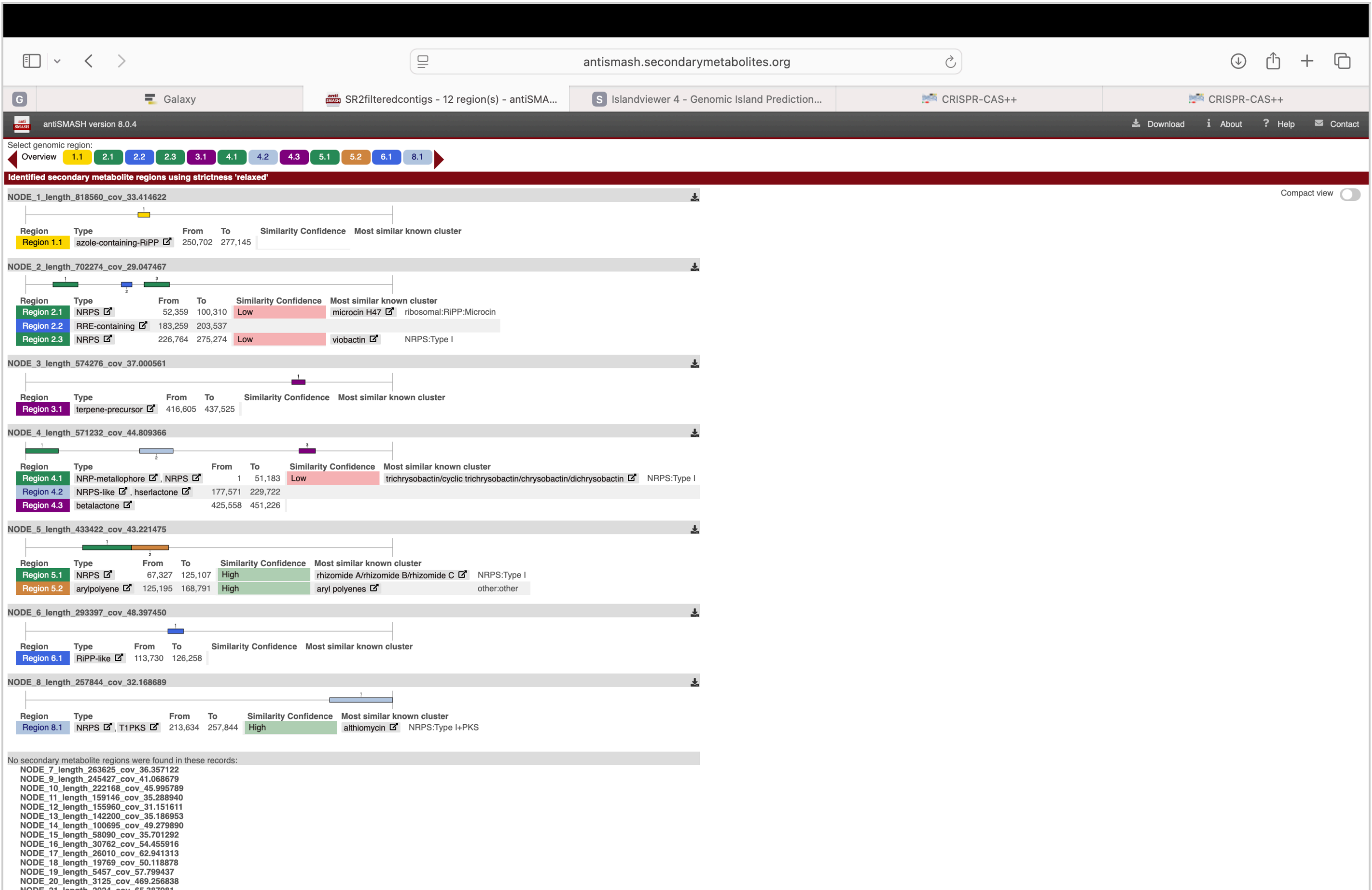
antiSMASH version 8.0.4

11E2. Key BGC finding(s) (2–4 sentences). Include at least one cluster type and whether it matches a known cluster:

AntiSMASH identified NRPS cluster located on node 5, which shows a high similarity confidence to the known rhizomide A/rhizomide B/thizomide C gene cluster. This finding is significant because NRPS clusters are often responsible for producing complex secondary metabolites, such as specialized antibiotics or signaling molecules.

Upload screenshot: antiSMASH results

Choose File SR2antiSM…sults.png



Save as `SampleID_antismash.png` in `miningfiles/` .

11F) IslandViewer (genomic islands)

Open: IslandViewer (https://www.pathogenomics.sfu.ca/islandviewer/)

11F1. IslandViewer version/database info (copy from results page if shown):

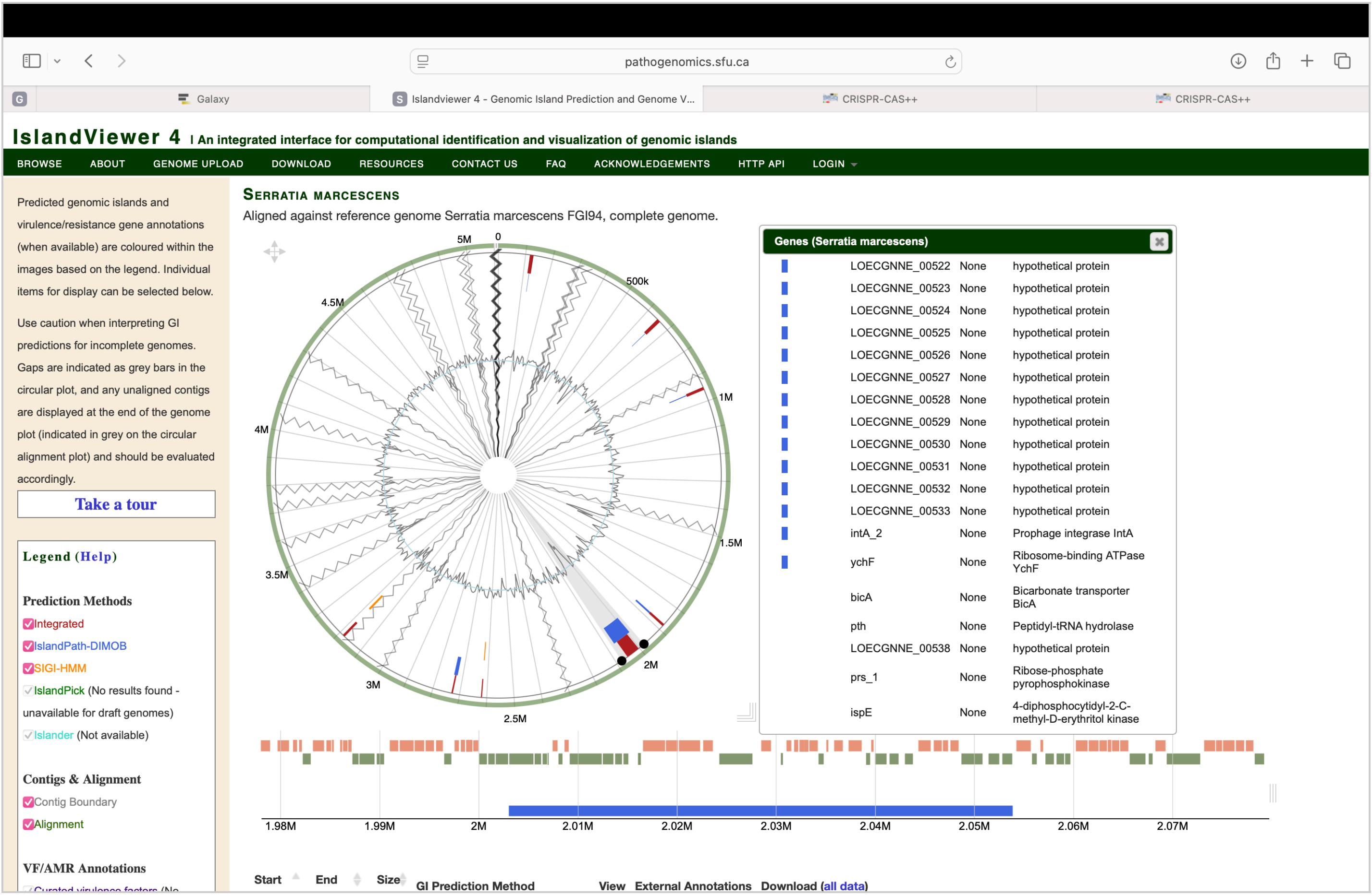
IslandViewer4

11F2. Key island finding(s) (2–4 sentences). How many predicted islands? Any notable features (mobility genes, tRNA insertion sites, etc.)?

There are 2 notable predicted islands. There is a gene in one island called intA_2 that is prophage. This indicates that the island was likely inserted into the bacterial genome by a virus.

Upload screenshot: IslandViewer results

Choose File SR2IslandV…sults.png



Save as `SampleID_islandviewer.png` in `miningfiles/` .

11G) CRISPRCasFinder (CRISPR arrays + Cas genes)

Open: CRISPRCasFinder (https://crisprcas.i2bc.paris-saclay.fr/)

11G1. CRISPRCasFinder version/database info (copy from results page if shown):

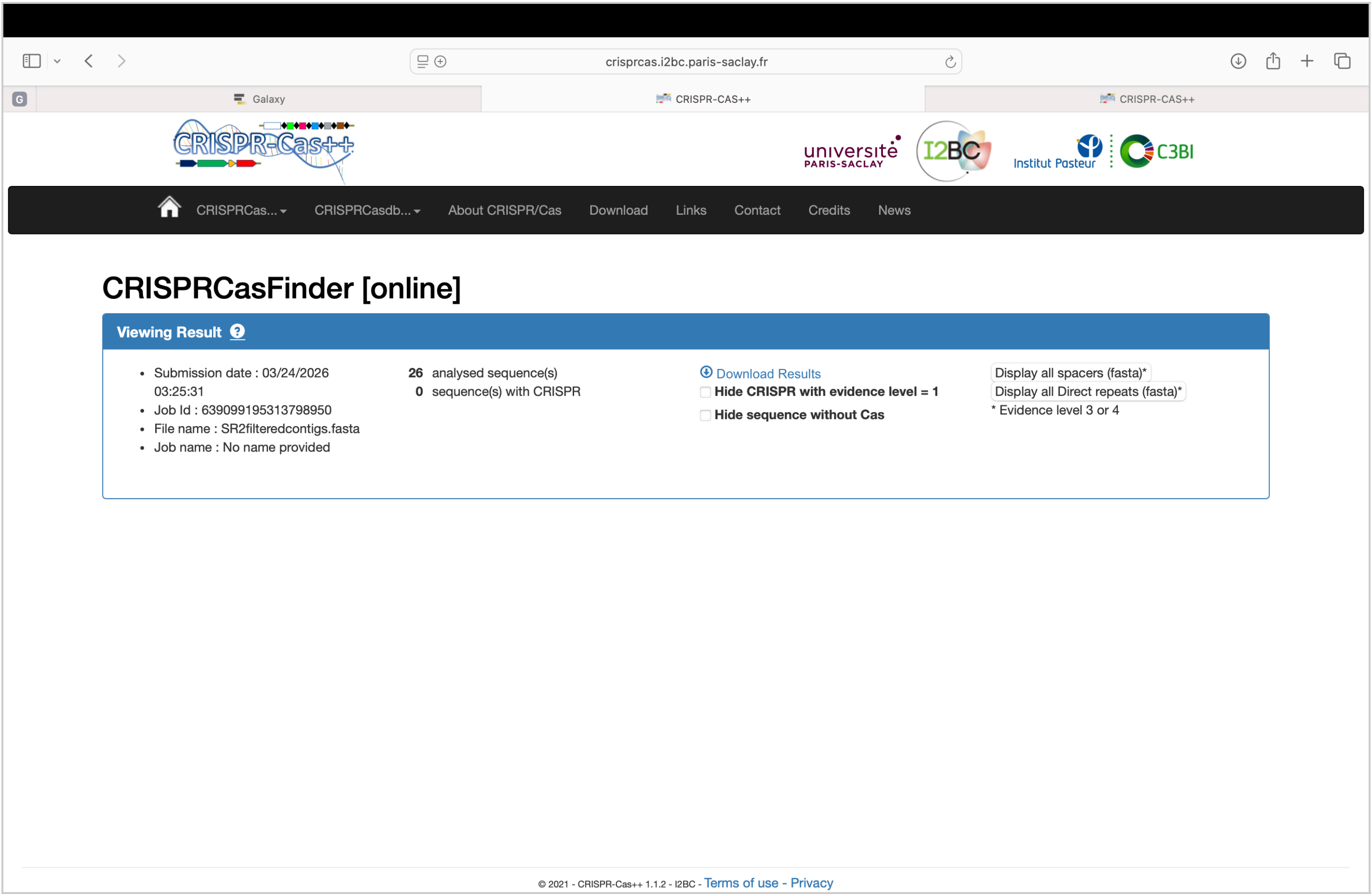
CRISPRCasFinder [online]

11G2. Key CRISPR/Cas findings (2–4 sentences). Did you find CRISPR arrays? Cas genes? What confidence level?

There were no CRISPR/ CAS findings. This lack of findings means that the bacteria lacks this specific adaptive immune system used to defend against viral infections.

Upload screenshot: CRISPRCasFinder results

Choose File SR2CRISPR…ults.png



Save as `SampleID_crisprcasfinder.png` in `miningfiles/` .

11H) Synthesis across genome mining tools

11H1. In 6–10 sentences, synthesize what genome mining suggests about this organism (resistance potential, virulence potential, mobile elements, secondary metabolism, etc.). Point to at least one result from each tool:

Genome mining suggests this organism is adaptable and a human pathogen. PathogenFinder confirms this with a high probability score (0.9707). ResFinder and CARD both identified the aac(6 ')–Ic gene, which is a specific resistance marker. CARD also identifies the adeF and SRT–4 genes under strict criteria. The adeF gene is part of an efflux pump system that physically moves antibiotics out of the cell, while SRT–4 is a beta-lactamase that can break down penicillin-like drugs. ABRicate further identifies virulence factors like flgH, fliM, and fliG, which are components of a flagellum that allow the bacteria to move and colonize hosts. The organism lacks a CRISPR immune system. IslandViewer shows the "genomic islands" has the intA_2 prophage gene. Finally, antiSMASH shows the NRPS cluster on node 5 that produces rhizomide-like

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12) Final synthesis (your conclusion)

12A. What is your best-supported species ID? Use at least 3 evidence types (16S, ANI, TYGS, CheckM2, Prokka) and explain remaining uncertainty (8–12 sentences):

My best-supported species ID for this organism is Serratia marcescens. ANI (Average Nucleotide Identity) analysis from EZBioCloud helped me reach this conclusion. The typical species-level match requires an ANI value >95%, and my results aligned with S. marcescens strains. 16S rRNA sequencing was performed, but some uncertainty remains because the Serratia species often share similarity in their 16S genes, making it difficult to distinguish S. marcescens from other close relatives. CheckM2 was used to see the high completeness and low contamination, so we can see that it is a pure bacterial genome. The Blastn results helped me by matching my isolate to S. marcescens in the NCBI database. Despite the previous evidence, there is still some uncertainty remains. The lack of CRISPR arrays raised a red flag. This means that my bacteria lacks the arrays that are susceptible to viral infections. They dont have any "molecular memory" to fight them off.

Export as PDF

When you are ready: 1. Ensure your screenshots/PDFs are uploaded into the page. 2. Click **Download as PDF (Print)** and save the PDF. 3. Verify that your notes file and your PDF are in the folder to be uploaded. 4. Commit your **folder structure + downloaded reports + renamed files + notes** to Git.